Worksheet 39 · Area & Perimeter IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Find the area and perimeter of 2D shapes • Find the area of trapeziums, parallelograms and compound shapes • Find the area and circumference of circles		
Grade: / 84	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Area & Perimeter*

Question 1 (4 marks)

The diagram shows a shape.

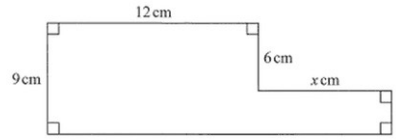


Diagram **NOT** accurately drawn

The shape has area 129 cm^2

Work out the value of x.

$x =$ .....

Question 2 (4 marks)

Here is a hexagon $ABCDEF$.

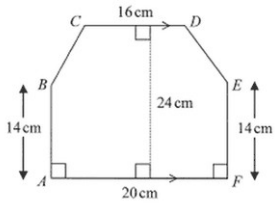


Diagram **NOT** accurately drawn

CD is parallel to AF.

Work out the area of hexagon $ABCDEF$.

..... cm^2

Question 3 (3 marks)

The diagram shows Yuen's garden.

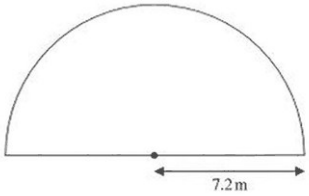


Diagram **NOT** accurately drawn

The garden is in the shape of a semicircle of radius 7.2 m.
Yuen is going to cover his garden with grass seed.

Yuen has 12 boxes of grass seed.
Each box of grass seed contains enough seed to cover 6 m^2 of the garden.

Has Yuen enough grass seed for his garden?
Show your working clearly.

Question 4 (3 marks)

The shape, shown shaded in the diagram, is the region between two semicircles.

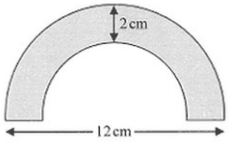


Diagram **NOT** accurately drawn

The diameter of the outer semicircle is 12 cm.
The shape has constant thickness 2 cm.

Calculate the area of the shape.
Give your answer as a multiple of π.

..... cm^2

Question 5 (5 marks)

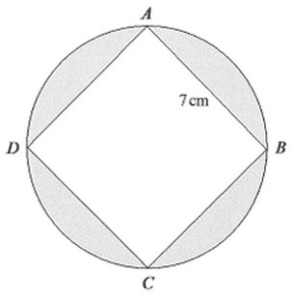


Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle.
ABCD is a square of side 7 cm.

Work out the total area of the shaded regions.
Give your answer correct to the nearest whole number.

..... cm^2

Question 6 (4 marks)

The diagram shows an 8-sided shape *ABCDEFGH*.

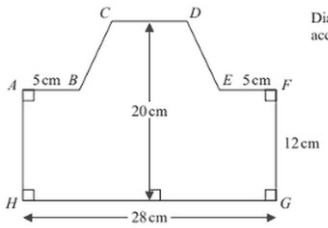


Diagram **NOT** accurately drawn

$HG = 28$ cm $FG = 12$ cm $AB = EF = 5$ cm
The height of the shape is 20 cm
CD is parallel to *HG*

The area of shape *ABCDEFGH* is 434 cm^2

Find the length of *CD*.

..... cm

Question 7 (3 marks)

The diagram shows a rectangle *ABCD* and a semicircle with diameter *AB* where *AB* = 12 cm. The point *E* lies on *DC* and also on the semicircle.

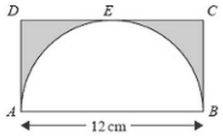


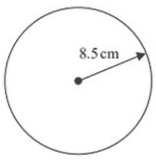
Diagram **NOT** accurately drawn

Work out the area of the shaded region.
Give your answer correct to 3 significant figures.

..... cm^2

Question 8 (4 marks)

The diagram shows a circle and a trapezium.



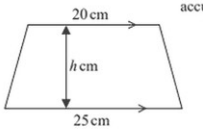


Diagram **NOT** accurately drawn

The height of the trapezium is *h* cm.

The area of the circle is equal to the area of the trapezium.

Work out the value of *h*.
Give your answer correct to 1 decimal place.

.....

Question 9 (32 marks)

The region, shown shaded in the diagram, is a path.

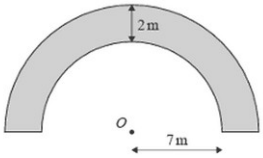


Diagram **NOT** accurately drawn

The boundary of the path is formed by two semicircles, with the same centre *O*, and two straight lines.

The inner semicircle has a radius of 7 metres.
The path has a width of 2 metres.

Work out the perimeter of the path.
Give your answer correct to one decimal place.

..... m

Question 10 (4 marks)

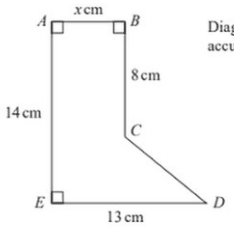


Diagram **NOT** accurately drawn

The diagram shows the shape *ABCDE*.

The area of the shape is 91.8 cm^2

Work out the value of *x*.

x = ...

Question 11 (5 marks)

The diagram shows the front of a wooden door with a semicircular glass window.

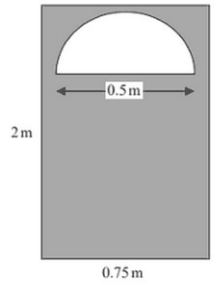


Diagram **NOT** accurately drawn

Julie wants to apply 2 coats of wood varnish to the front of the door, shown shaded in the diagram.

250 millilitres of wood varnish covers 4 m^2 of the wood.

Work out how many millilitres of wood varnish Julie will need.
Give your answer correct to the nearest millilitre.

..... millilitres

Question 12 (4 marks)

The diagram shows a shape made from a square *ABCD* and 4 identical semicircles.

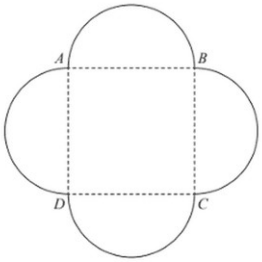


Diagram **NOT** accurately drawn

As shown in the diagram, the semicircles have *AB*, *BC*, *CD* and *DA* as diameters.

The area of the square is 36 cm^2

Calculate the total area of the shape.
Give your answer correct to one decimal place.

..... cm^2

Question 13 (5 marks)

The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.

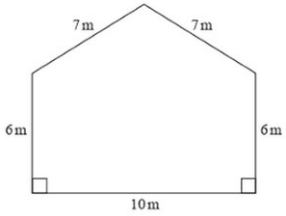


Diagram **NOT** accurately drawn

Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

The paint in each tin of paint Omondi is going to buy will cover 16 m^2 of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

.....

Question 14 (4 marks)

The diagram shows four identical circles drawn inside a square.

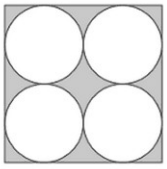


Diagram **NOT** accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.
Give your answer correct to 3 significant figures.

..... cm